



TECH ORBIT

Full Stack MERN + Agentic AI Program

"From Zero to Full Stack AI Engineer"

India's First Comprehensive Full Stack + Agentic AI Program

Master the complete technology stack—from building scalable web applications with MongoDB, Express, React, and Node.js to creating autonomous AI agents that can reason, plan, and execute complex tasks.

A 100% practical, hands-on curriculum designed to make you a complete Full Stack AI Engineer ready for the jobs of tomorrow.



MongoDB



Express.js



React



Node.js



Agentic AI

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What Makes Tech Orbit Different?

Two powerful skill tracks in one comprehensive program



Full Stack MERN

Build production-ready web applications using the most in-demand JavaScript stack.

- React.js with modern hooks & state management
- Node.js & Express.js backend development
- MongoDB database design & optimization | RESTful APIs



Agentic AI

Create autonomous AI systems that can understand, reason, and take action.

- LLM integration (GPT-4, Claude, Gemini)
- RAG systems & vector databases
- AI agents with planning & execution
- LangChain, LangGraph & production AI

The Complete Package

Most programs teach either web development OR AI—we teach you both and show you how to combine them. Build intelligent web applications where your React frontend connects to AI-powered backends that can understand natural language, process documents, and make decisions autonomously.

100% Hands-On Learning

Every concept taught through building real projects. No theory-heavy lectures—you learn by creating actual applications.

Offline Learning Centers

Face-to-face instruction with expert mentors available to help you immediately when you're stuck.

Industry-Ready Portfolio

Graduate with 20+ real projects that employers want to see—including full stack apps with AI integration.

Integrated Curriculum

Learn how to combine MERN stack with AI capabilities to build the intelligent applications companies need.

Who Is This Program For?

- **College Students:** Engineering students wanting to build dual expertise in web development and AI
- **Fresh Graduates:** Looking to stand out with both full stack and AI skills
- **Career Switchers:** Professionals wanting to transition into high-demand tech roles
- **Web Developers:** Existing developers wanting to add AI capabilities to their toolkit
- **Complete Beginners:** Anyone with zero experience ready to become a full stack AI engineer

Our Promise

By the end of this program, you'll have built 20+ real projects spanning full stack applications and AI systems—ready for roles at top tech companies.

Complete Curriculum Overview

A structured journey from fundamentals to advanced full stack AI engineering

Phase 1: Programming Foundations

1. Python Fundamentals

What You'll Build: Smart applications including a thermostat system, delivery fee calculator, and automation tools

- Installation & Setup (VSCode, Python, Virtual Environments)
- Data Types: Numbers, Strings, Lists, Tuples, Sets, Dictionaries
- Conditionals, Loops, and Functions
- Object-Oriented Programming: Classes, Inheritance, Design Patterns
- Error Handling and File Operations
- Comprehensions, Generators & Decorators

2. JavaScript Fundamentals

What You'll Build: Interactive web applications and dynamic user interfaces

- ES6+ Syntax: let/const, arrow functions, template literals
- Data Structures: Arrays, Objects, Maps, Sets
- DOM Manipulation and Event Handling
- Asynchronous JavaScript: Callbacks, Promises, Async/Await
- Closures, Prototypes, and the 'this' keyword
- Modern JS: Destructuring, Spread/Rest, Modules

3. Advanced Programming Concepts

Master professional development techniques

- Concurrency: Threading, Multiprocessing, AsyncIO (Python)
- Event Loop and Non-blocking I/O (JavaScript)
- Data Validation with Pydantic and Zod
- TypeScript fundamentals for type-safe JavaScript
- Design Patterns and Clean Code principles

Phase 2: MERN Stack Development

4. React.js Frontend Development

Project: Build a modern, responsive web application

- React fundamentals: Components, JSX, Props
- State Management: useState, useReducer, Context API
- React Hooks: useEffect, useRef, useMemo, useCallback, Custom Hooks
- Routing with React Router v6
- Forms and Validation (React Hook Form, Formik)
- Styling: Tailwind CSS, Styled Components, CSS Modules
- State Management: Redux Toolkit, Zustand
- Performance Optimization and Code Splitting

5. Node.js & Express.js Backend

Project: Build scalable REST APIs and server applications

- Node.js runtime and npm ecosystem
- Express.js: Routing, Middleware, Error Handling
- RESTful API Design best practices
- Authentication: JWT, OAuth, Passport.js
- Input Validation and Sanitization
- File Uploads and Streaming
- Rate Limiting and Security Best Practices
- API Documentation with Swagger/OpenAPI

6. MongoDB Database

Project: Design and implement production-ready databases

- MongoDB fundamentals and document model
- CRUD Operations and Query Optimization
- Mongoose ODM: Schemas, Models, Validation
- Relationships: Embedding vs. Referencing
- Aggregation Pipeline for complex queries
- Indexing strategies for performance
- MongoDB Atlas: Cloud deployment
- Data backup and migration strategies

7. Full Stack Integration

Project: Build a complete production application

- Connecting React frontend to Express backend
- Environment configuration and secrets management
- CORS, Proxies, and API integration patterns
- Real-time features with Socket.io
- GraphQL with Apollo (Server & Client)
- Testing: Jest, React Testing Library, Supertest
- Error tracking and logging (Winston, Morgan)

Phase 3: AI & LLM Foundations

8. Understanding Large Language Models

Deep dive into the technology powering ChatGPT, Claude, and Gemini

- How LLMs work under the hood
- GPT Architecture and Transformer fundamentals
- Tokenization: How AI reads and understands text
- Vector Embeddings: The mathematics of meaning
- Multi-Head Attention: How AI maintains context
- Model parameters, context windows, and limitations

9. LLM API Integration

Hands-on: Connect to and use real AI models

- OpenAI API: GPT-4, GPT-4o integration
- Anthropic Claude API setup and usage
- Google Gemini integration
- API best practices: Rate limiting, error handling, cost optimization
- Building your first AI-powered application
- Streaming responses for better UX

10. Advanced Prompt Engineering

Master the art of communicating with AI

- Zero-Shot, One-Shot, Few-Shot Prompting
- Chain-of-Thought (CoT) Reasoning
- Persona-Based Prompting for specific use cases
- Structured Outputs: Getting AI to return JSON, XML, etc.
- Prompt chaining and decomposition
- Prompt templates and management

11. Running AI Models Locally

Deploy powerful AI without external API dependency

- Ollama: Local LLM runtime engine
- Docker setup for AI development
- OpenWebUI for local model interfaces
- Hugging Face Hub: Access to open-source models
- Model quantization and optimization
- When to use local vs. cloud models

Phase 4: Agentic AI Systems

12. Building AI Agents

Create AI that can plan, reason, and execute tasks autonomously

- What are AI Agents? Core concepts and architecture
- Coding your first AI agent from scratch
- Tool use: Giving agents abilities to take action
- Structured outputs with Pydantic validation
- Building a CLI coding agent (like Claude Code)
- Agent decision-making and action execution

13. RAG Systems (Retrieval-Augmented Generation)

Project: Build "Chat with PDF" – Make AI understand your documents

- Understanding the RAG problem and solution
- LangChain framework mastery
- Document loading and processing (PDF, DOCX, TXT)
- Text chunking strategies for optimal retrieval
- Vector databases: Pinecone, ChromaDB, Qdrant
- Embedding models and similarity search
- Building the complete RAG pipeline
- Evaluation and improving RAG quality

14. LangGraph: Complex AI Workflows

Build sophisticated AI systems with multiple reasoning steps

- Why LangGraph is revolutionary for AI agents

- Nodes and Edges: Designing AI decision graphs
- State management in AI workflows
- Conditional routing: Smart decision-making
- Human-in-the-loop patterns
- Checkpointing with MongoDB: Never lose progress
- Testing and debugging complex AI graphs

15. Memory Systems for AI

Build AI that learns from every interaction

- Short-Term Memory: Conversation context
- Long-Term Memory: Persistent knowledge across sessions
- Episodic Memory: Remembering specific events
- Semantic Memory: General knowledge storage
- Mem0: Modern memory management for AI
- Knowledge Graphs with Neo4j

Phase 5: Full Stack AI Integration

16. AI-Powered MERN Applications

**Project: Build intelligent web applications combining
MERN + AI**

- Integrating LLM APIs with Express.js backends
- Building AI chat interfaces in React
- Storing conversations and context in MongoDB
- Real-time AI responses with Socket.io
- Building AI-powered search and recommendations
- Content generation and moderation systems

17. Multi-Modal AI Applications

AI that understands images, audio, and video—not just text

- Image processing with Vision models
- Speech-to-Text (STT) and Text-to-Speech (TTS)
- Building voice-controlled interfaces
- Video analysis and processing
- Multi-modal RAG systems

18. Scalable AI Architecture

Build AI systems that handle production traffic

- Async processing with message queues (Redis, BullMQ)
- Worker architecture for AI tasks
- Caching strategies for AI responses
- Load balancing AI workloads
- Cost optimization for API calls

- Monitoring and observability

19. Model Context Protocol (MCP)

Latest: Standardized AI-tool communication

- Understanding MCP architecture
- Building MCP servers and clients
- Integrating external tools with AI agents
- Real-world MCP implementations

Phase 6: DevOps & Deployment

20. Git Version Control

Essential for every developer—manage your code like a pro

- Git fundamentals: init, commits, branches, merges
- Branching strategies (GitFlow, Trunk-based)
- Resolving merge conflicts
- GitHub collaboration: PRs, code reviews, issues
- Git hooks and automation
- Open-source contribution workflow

21. Docker & Containerization

Deploy your applications anywhere with confidence

- Containerization fundamentals
- Writing optimized Dockerfiles
- Multi-stage builds for smaller images
- Docker Compose for multi-container apps
- Container networking and volumes
- Containerizing MERN + AI applications

22. Cloud Deployment

Production deployment on major cloud platforms

- AWS: EC2, ECS, ECR, S3, RDS
- Vercel deployment for React applications
- Railway/Render for backend services
- MongoDB Atlas for database hosting
- CI/CD with GitHub Actions
- Environment management and secrets
- SSL, domains, and DNS configuration
- Cost optimization and monitoring

Capstone Projects

Build these production-ready applications for your portfolio

1. E-Commerce Platform with AI

Full MERN e-commerce with AI-powered product recommendations, search, and customer support chatbot

2. Document Intelligence System

Upload documents and chat with them using RAG. Built with React frontend and AI-powered backend

3. AI Coding Assistant

Build your own Claude Code-like assistant that can understand codebases and help developers

4. Multi-Agent Workflow System

Create a system where multiple AI agents collaborate to complete complex tasks

5. Real-time Collaboration App

Build a Notion-like app with real-time editing, AI writing assistance, and smart organization

6. Voice-Controlled Dashboard

Analytics dashboard with voice commands, natural language queries, and AI-generated insights

Career Outcomes

20+

Real Projects Built

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Skill Tracks Mastered

22

Comprehensive Modules

100%

Hands-On Learning

You'll Be Ready For

✓ Full Stack Developer / MERN Developer roles

✓ AI Engineer / ML Engineer positions

✓ Full Stack AI Engineer roles (the most in-demand hybrid role)

✓ Freelancing and consulting on both web and AI projects

Launch Your Full Stack AI Career

Join India's first comprehensive Full Stack MERN + Agentic AI program.
Transform from complete beginner to job-ready Full Stack AI Engineer
in our offline learning centers.



Email Us

info@techorbitindia.com
Response within 24 hours



Call Us

+91 91774 77855
Mon-Sat: 10AM - 7PM



Visit Us

Unit 201, 2nd Floor, Siri Plaza
KPHB, Hyderabad - 500072

Why Choose Tech Orbit?

We're building the next generation of Full Stack AI Engineers—developers who can build complete applications from database to UI AND integrate intelligent AI capabilities throughout.



Dual Expertise - Master both MERN stack and Agentic AI in one program



Integrated Curriculum - Learn to combine web development with AI



Offline Classrooms - Face-to-face mentorship and guidance



Career-Focused - Skills that make you stand out in the job market

Full Stack. AI-Powered. Career-Ready.

 **Enroll Now**